

## HONEFOSS-HOYBY, Norway

WMO: 014690

Lat: 60.1660N

Lon: 10.2463E

Elev: 140

StdP: 99.65

Time Zone: 1.00 (EUC)

Period: 05-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -17.5      | -15.2 | -19.9                         | 0.7 | -17.3 | -17.6 | 0.8 | -14.5 | 7.0                   | 2.1  | 6.0 | 1.9  | 1.5                   | 340  | 0.566 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.9                    | 27.5            | 16.9 | 25.5 | 16.2 | 23.3 | 15.3 | 18.2                | 24.7 | 17.3 | 23.1 | 16.4 | 21.4 | 2.5                  | 160  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 16.0                            | 11.5 | 20.0 | 15.1 | 10.9 | 19.3 | 14.2 | 10.3 | 18.5 | 52.0          | 24.2 | 49.2 | 23.0 | 46.4 | 21.4 | 21.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
| (4) 6.3           | 5.4  | 4.7 | -18.5                      | 29.9 | 3.5                | 2.0 | -21.1                                              | 31.3 | -23.1      | 32.5 | -25.1      | 33.6 | -27.6      | 35.0 |  |  |
| (5)               |      |     | -18.8                      | 19.5 | 3.6                | 1.0 | -21.4                                              | 20.2 | -23.5      | 20.8 | -25.5      | 21.3 | -28.1      | 22.1 |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 6.2    | -4.9  | -3.6  | 0.8   | 5.9   | 11.0  | 15.1  | 17.7  | 15.8  | 11.9  | 5.8   | 1.0   | -3.2  |
|                                                                        | DBStd       | 8.74   | 5.39  | 5.18  | 4.48  | 3.36  | 3.98  | 2.84  | 2.91  | 2.28  | 2.97  | 3.62  | 4.05  | 5.69  |
|                                                                        | HDD10.0     | 2127   | 462   | 381   | 286   | 129   | 34    | 1     | 0     | 0     | 16    | 137   | 271   | 409   |
|                                                                        | HDD18.3     | 4488   | 721   | 614   | 544   | 373   | 229   | 106   | 49    | 83    | 193   | 388   | 521   | 667   |
|                                                                        | CDD10.0     | 723    | 0     | 0     | 1     | 7     | 66    | 152   | 238   | 180   | 72    | 8     | 1     | 0     |
|                                                                        | CDD18.3     | 43     | 0     | 0     | 0     | 0     | 2     | 7     | 29    | 5     | 0     | 0     | 0     | 0     |
|                                                                        | CDH23.3     | 436    | 0     | 0     | 0     | 0     | 40    | 104   | 262   | 29    | 0     | 0     | 0     | 0     |
|                                                                        | CDH26.7     | 81     | 0     | 0     | 0     | 0     | 4     | 18    | 59    | 1     | 0     | 0     | 0     | 0     |
| (14) Wind                                                              | WSAvg       | 2.2    | 2.1   | 2.1   | 2.3   | 2.3   | 2.4   | 2.3   | 2.1   | 2.0   | 2.1   | 2.1   | 2.2   | 2.0   |
| (15) Precipitation                                                     | PrecAvg     | 771    | 54    | 36    | 38    | 46    | 62    | 76    | 87    | 90    | 77    | 82    | 77    | 48    |
|                                                                        | PrecMax     | 1115   | 113   | 87    | 115   | 95    | 124   | 152   | 160   | 177   | 186   | 203   | 312   | 80    |
|                                                                        | PrecMin     | 594    | 13    | 11    | 0     | 7     | 8     | 11    | 16    | 27    | 30    | 19    | 13    | 11    |
|                                                                        | PrecStd     | 101    | 25    | 19    | 24    | 24    | 30    | 38    | 30    | 35    | 34    | 42    | 55    | 20    |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 7.6   | 9.0   | 15.1  | 19.6  | 27.1  | 28.9  | 30.2  | 26.0  | 22.3  | 16.4  | 12.1  | 9.5   |
|                                                                        |             | MCWB   | 4.9   | 4.3   | 8.0   | 10.9  | 15.4  | 17.1  | 18.3  | 16.9  | 15.2  | 12.4  | 9.1   | 6.4   |
|                                                                        | 2%          | DB     | 5.5   | 6.5   | 12.1  | 17.5  | 24.0  | 26.5  | 28.4  | 24.0  | 20.2  | 14.3  | 10.3  | 7.7   |
|                                                                        |             | MCWB   | 3.2   | 3.2   | 6.2   | 9.9   | 14.2  | 15.6  | 17.9  | 16.1  | 13.9  | 10.7  | 7.7   | 5.5   |
|                                                                        | 5%          | DB     | 3.7   | 4.7   | 9.2   | 15.1  | 21.4  | 23.9  | 26.5  | 22.2  | 18.6  | 13.0  | 8.6   | 6.1   |
|                                                                        |             | MCWB   | 1.9   | 2.4   | 4.5   | 8.5   | 12.8  | 14.5  | 17.4  | 15.5  | 13.4  | 10.0  | 6.6   | 3.8   |
|                                                                        | 10%         | DB     | 2.3   | 3.1   | 7.0   | 12.6  | 18.8  | 21.5  | 24.2  | 20.6  | 16.9  | 11.5  | 7.0   | 4.4   |
|                                                                        |             | MCWB   | 0.8   | 1.5   | 3.1   | 6.8   | 12.0  | 13.5  | 16.2  | 14.7  | 12.6  | 9.0   | 5.4   | 2.5   |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 5.2   | 5.1   | 8.4   | 11.9  | 16.9  | 17.7  | 20.0  | 18.3  | 16.4  | 13.4  | 9.9   | 7.2   |
|                                                                        |             | MCDB   | 7.2   | 7.2   | 14.8  | 17.8  | 23.9  | 26.1  | 27.1  | 22.8  | 19.8  | 15.2  | 11.6  | 9.1   |
|                                                                        | 2%          | WB     | 3.7   | 3.7   | 6.5   | 10.4  | 15.2  | 16.5  | 18.9  | 17.4  | 15.1  | 11.5  | 8.1   | 5.6   |
|                                                                        |             | MCDB   | 5.2   | 5.8   | 11.5  | 16.2  | 21.9  | 23.9  | 26.2  | 21.6  | 18.4  | 13.5  | 9.9   | 7.5   |
|                                                                        | 5%          | WB     | 2.1   | 2.6   | 4.9   | 9.0   | 13.8  | 15.6  | 18.0  | 16.5  | 14.2  | 10.4  | 6.8   | 4.1   |
|                                                                        |             | MCDB   | 3.5   | 4.4   | 8.3   | 13.8  | 20.0  | 21.7  | 24.7  | 20.4  | 17.5  | 12.4  | 8.3   | 5.9   |
|                                                                        | 10%         | WB     | 0.9   | 1.4   | 3.7   | 7.6   | 12.3  | 14.5  | 17.1  | 15.7  | 13.2  | 9.3   | 5.5   | 2.6   |
|                                                                        |             | MCDB   | 2.3   | 3.0   | 6.5   | 12.1  | 17.2  | 19.9  | 22.9  | 19.3  | 16.1  | 11.3  | 7.1   | 4.1   |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 4.9   | 5.9   | 8.5   | 10.1  | 10.3  | 10.1  | 9.9   | 9.0   | 8.6   | 6.5   | 4.9   | 4.9   |
|                                                                        |             | MCDBR  | 6.2   | 8.1   | 11.8  | 14.2  | 14.7  | 14.4  | 13.5  | 11.2  | 10.9  | 7.8   | 7.3   | 6.9   |
|                                                                        |             | MCWBR  | 5.1   | 5.6   | 7.2   | 7.9   | 6.9   | 6.3   | 5.3   | 5.1   | 6.1   | 5.2   | 5.9   | 5.8   |
|                                                                        | 5% WB       | MCDBR  | 6.0   | 7.0   | 10.7  | 13.2  | 12.5  | 11.9  | 12.0  | 9.2   | 9.0   | 6.6   | 6.8   | 6.7   |
|                                                                        |             | MCWBR  | 5.3   | 5.1   | 6.8   | 7.6   | 6.2   | 5.9   | 5.2   | 4.7   | 5.4   | 5.1   | 5.9   | 5.8   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.245 | 0.288 | 0.317 | 0.360 | 0.360 | 0.351 | 0.371 | 0.363 | 0.330 | 0.315 | 0.266 | 0.216 |
|                                                                        | taud        |        | 2.198 | 2.285 | 2.310 | 2.320 | 2.420 | 2.467 | 2.429 | 2.445 | 2.525 | 2.483 | 2.292 | 2.102 |
|                                                                        | Ebn at Noon |        | 543   | 718   | 813   | 837   | 868   | 881   | 851   | 829   | 804   | 691   | 515   | 418   |
|                                                                        | Edh at Noon |        | 39    | 65    | 88    | 105   | 102   | 99    | 101   | 92    | 72    | 55    | 36    | 27    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 0.29  | 0.92  | 2.40  | 3.84  | 4.83  | 5.48  | 4.89  | 3.85  | 2.54  | 1.14  | 0.38  | 0.19  |
|                                                                        | RadStd      |        | 0.05  | 0.18  | 0.24  | 0.52  | 0.51  | 0.44  | 0.64  | 0.32  | 0.31  | 0.22  | 0.10  | 0.02  |

## Historical Trends

|                              | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (46) Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (57 neighbors) | N/A   | N/A     | +1.56  | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page